3 **(a)** A car has a weight of 13000 N. The car is supported by 4 tyres. The area of each tyre in contact with the road is 0.016 m^2.

(i) Calculate the pressure on the road due to the weight of the car.

pressure = ... [2]

(ii) Explain, in terms of particles, why the air pressure in the tyres increases when the car travels along the road.

.....

.....

.....

.....

..... [4]

(b) A gas cylinder contains helium gas at a pressure of 2.0×10^6 Pa. A volume of 0.026 m^3 of the compressed gas is released from the cylinder into balloons. Each balloon contains 0.015 m^3 of helium at atmospheric pressure (1.0×10^5 Pa). The temperature remains constant.

Calculate the maximum number of balloons that can be filled.

maximum number of balloons = ... [3]

[Total: 9]

© UCLES 2024